NYU Center for Data Science

Budget-Aware PPO for Real-Time Ad Bidding

Investigation of training selective PPO to learn budget-aware bidding strategies in large-scale, budget constrained auction environments

Team BEAS

Background & Problem

Real Time Bidding (RTB)

In modern online advertising, most ad impressions are sold through Real-Time Bidding (RTB). This happens when a user visits a webpage or opens an app:

- An auction is triggered in milliseconds
- Multiple advertisers submit bids
- The highest bidder wins and shows their ad

This process is a sequential decision-making problem.

Core Challenges

1. Budget Constraint
2. Uncertainty
    a. Not knowing whether or not a user would convert
    b. Reliance on predicted probabilities
3. Sparse and Delayed Reward
    a. Many impressions do not lead to conversion
    b. Reward signals are rare

Can on-policy PPO learn a better budget-aware bidding strategy than fixed bidding rules and AuctionNet offline-RL baselines?

AuctionNet Environment

AuctionNet provides an environment that effectively replicates the integrity and complexity of real-world ad auctions with the interaction of the ad opportunity generation module, the bidding module, and the auction module.

Episode Structure

- One episode = one delivery period
- 48 decision ticks per episode
- Each tick contains multiple ad opportunities

Auction Setting

- Agent competes with ~47 other bidders
- Second-price auction mechanism
- Win if bid ≥ market price

Decision Flow

- Observe state (pValue, budget, time)
- Output bid multiplier (policy)
- Participate in auction
- Receive cost & conversion signal
- Compute reward

Observation → policy → bid multiplier/ selectivity → auction → cost/conversions → reward

Key Evaluation Metrics

1	Conversions Total number of successful user actions generated by the agent.
2	Return of Interest (ROI) Total conversion value divided by total spend. ROI measures the **efficiency of spending**.
3	Budget Utilization Total budget spend divided by the total amount of available funds.
4	Win/Exposure Rate The ratio between the impressions won and the total number of bidding opportunities.
5	Cost per Acquisition (CPA) The ratio between total cost and the total number of conversions.

From Budget Sweep to Binding-Regime Identification

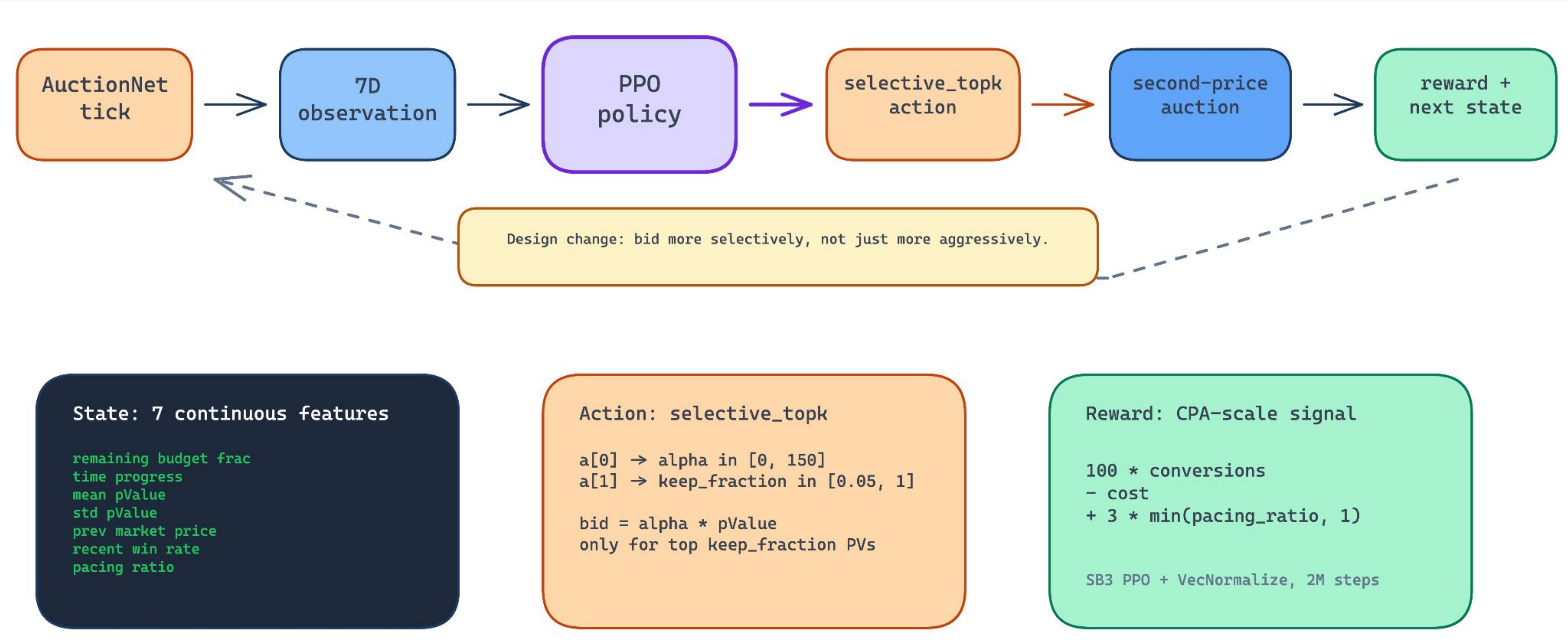
Budget = 2900 (Non-binding)

- Budget rarely spent
- Weak reward signal
- Utilization capped (~5%)
- PPO cannot learn meaningful policy

Budget = 150 (Binding)

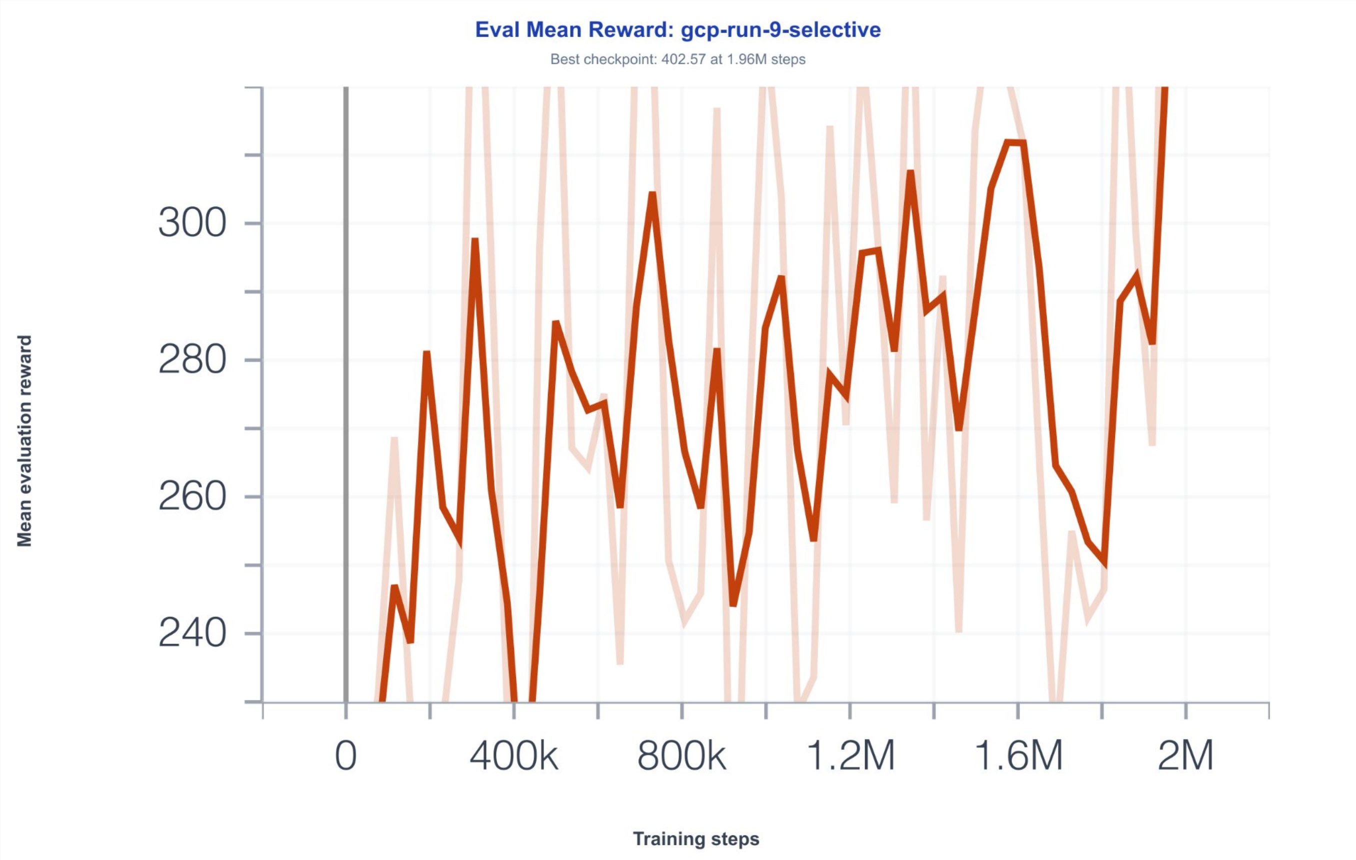
- Budget actively constraints bidding
- Clear ROI vs utilization tradeoff
- Policies already show structure
- Suitable for policy comparison

PPO Design: Turning Action into a Budget-Pacing MDP



Training Progress: Stable PPO

Critique learned meaningful structure, exploration did not collapse, and VecNormalize stats are saved and restored for checkpoint evaluation



Final Training Setup

- 2M timestamps
- 8 parallel envs
- [256, 256] policy/value networks
- lr = 1e-4
- ent_coef = 0.01
- Best eval reward: 402.57 at 1.96M steps
- Final explained variance: 0.861
- Final value loss: 0.0349
- Final policy std: 0.581
- Final approx KL: 0.0049

VecNormalized restored/saved

500-episode eval completed

PPO vs Fixed-Alpha Baselines

The table below shows the performance of PPO against fixed-alpha baselines

Policy	ROI	Util.	Win Rate	Reward
PPO selective	0.0273	66.64%	5.32%	299.60
α=100	0.0294	53.21%	4.87%	271.15
α=130	0.0234	69.96%	5.79%	266.17

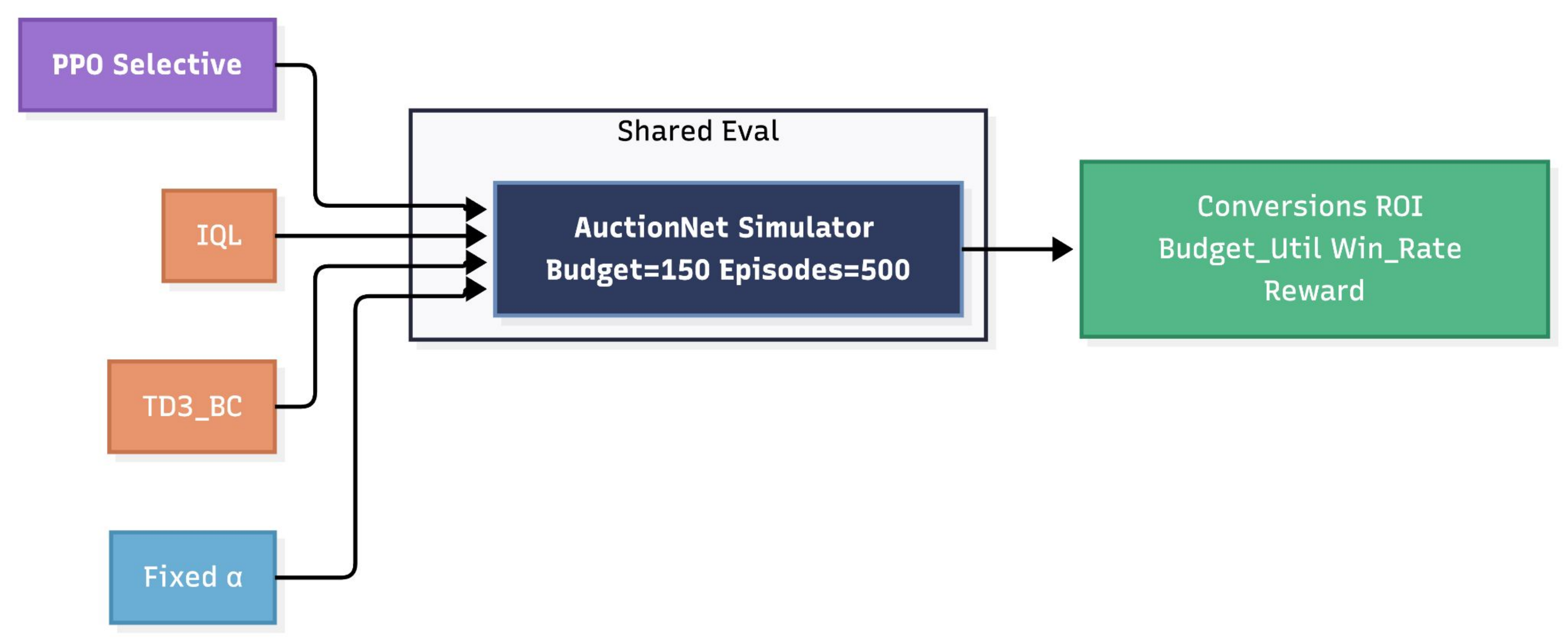
Fixed-alpha bid rule: `bid_t = α × pValue_t`

PPO reward: `R = 100 × conversions − cost + 3 × min(pacing_ratio, 1)`

In the fixed-alpha policy, we multipy every the pValue by a constant scalar α at every tick. α=100 bids 100× the pValue of every slot. α=130 bids 130×, which is more aggressive.

PPO achieved the best shaped reward and better overall reward/utilization tradeoff. Fixed α=100 remains slightly better on pure ROI, but spends less and gets lower reward.

Shared Evaluator for Offline RL Baselines



AuctionNet IQL does not natively report ROI in the same format as our PPO evaluator.
So we built a common evaluator that runs PPO, IQL, TD3_BC, and fixed alpha in one AuctionNet loop.

PPO vs AuctionNet Offline RL

The table below shows the performance of PPO against AuctionNet baselines

Policy	Conv.	Cost	Util.	CPA ↓	ROI ↑	Reward
PPO selective	1349	49,775	66.37%	36.90	0.02710	297.78
IQL	1140	74,554	99.41%	65.40	0.01529	162.06
Fixed α=100	1120	40,196	53.59%	35.89	0.02786	260.68
Fixed α=130	1203	52,987	70.65%	44.05	0.02270	259.57

IQL bid rule: `bid_t = α_t × pValue_t`

IQL reward: `R = ΣpValue_i`

PPO beats IQL on conversions, CPA, ROI, and shaped reward. IQL spends nearly all its budget but inefficiently i.e. budget utilization is 99.4 , but CPA is the worst.

Supporting TD3_BC Result

The table below includes the performance of TD3_BC

Policy	Conv.	Util.	CPA	ROI	Reward
PPO	129	66.17%	36.93	0.02708	296.76
IQL	109	99.38%	65.65	0.01523	161.87
TD3_BC	106	97.77%	66.41	0.01506	149.09

TD3_BC bid rule: `bid_t = α_t × pValue_t`

TD3_BC reward: `R = ΣpValue_i`

TD3_BC shows the same pattern as IQL: high spend, high exposure, poor efficiency.
This suggests the result is not specific to one offline-RL baseline.

Conclusion & Future Directions

1 **Environment & Setup**

- Budget regime critically determines learning quality
- Budget = 150 induces a binding, learnable pacing regime

2 **Policy Performance**

- Selective PPO achieves a stronger reward-utilization tradeoff
- PPO outperforms IQL and TD3_BC under aligned evaluation

3 **Nuance**

- PPO does not strictly maximize ROI; conservative policies may be more efficient via underspending.

Future work:

- Per-slot bidding: Instead of one α per tick, output a bid multiplier for each individual slot. The policy sees all pValues in a tick and learns to bid differently on each one based on its value, competition level, and remaining budget.
- Offline pre-training + PPO finetuning: pre-train a policy on the offline data to get a reasonable starting point, then fine-tune with PPO in simulator would compress training time significantly

THANK YOU